1986MNRAS.220..259L

Mon. Not. R. astr. Soc. (1986) **220,** 259–269

Cooling of Population III objects in a pressure supported collapse

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Accepted 1985 December 17. Received 1985 November 25; in original form 1985 August 29

Summary. The cooling and collapse of Population III objects is investigated in the framework of 'flat' standard cosmological model. We have assumed that the significant component of fluctuations was isothermal and that the cooling of the collapsing cloud was due to the H_2 molecule. We have taken into account the pressure inside the cloud. The only parameter in our models is the cloud's mass. Our numerical solutions show that the first bound systems were of about a million solar masses. In the range of 10^5–$10^7\,M_\odot$ there are four different modes of the behaviour of the cloud's radius, depending on the mass: expansion, expansion followed by collapse because of the cooling, collapse after oscillations due to the pressure, and a direct contraction. It is suggested that the different ways to collapse gave rise to different fragmentation processes and hence to different star systems with a similar mass. We find that the collapse was delayed by the pressure, and the first clouds completed their contraction at redshift 100–140.

1 Introduction

In this paper we reconsider the problem of the masses that first collapsed to form bound systems, the so-called Population III objects.

The formation of structures in the post-recombination gaseous medium is commonly explained by gravitational instabilities experienced by primordial density fluctuations. In a self-gravitating collapsing cloud the temperature tends to increase with contraction. A cooling mechanism is required to lower the pressure opposing gravity (Jeans criterion). The condition for efficient cooling is that the cooling time-scale be shorter than the free-fall time-scale (Yoneyama 1972). The difficulty in cooling the post-recombination clouds is due to the simplicity of the chemical composition of the cold medium. The main cooling agent is the hydrogen molecule that is formed during the collapse (Takayanagy & Nishimura 1960). According to Peebles & Dicke (1968) the first objects to form were the globular clusters. Recent works on that subject are by Hutchins (1976); Silk (1977, 1983); Carlberg (1981); Palla, Salpeter & Stahler (1983) and Lepp & Shull

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(1984). Most authors have neglected the pressure inside the cloud and have inserted the cosmological initial conditions by hand.

The innovations in the present work are two: we relate the local physical and chemical processes inside the collapsing cloud to the cosmological background in a way depending only on one free parameter – the mass of the cloud – we also take into account the pressure inside the cloud. In our numerical calculations we use new chemical reaction rates and a new cooling function.

We shall summarize our assumptions. We adopt the standard cosmological model with zero cosmological constant. The Hubble constant at present is written as $H_0=100\,h\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. In a matter dominated universe the density parameter Ω is close to 1 for redshift $z\gg\Omega_0^{-1}$ (where Ω_0 is the present density parameter) if $\Omega_0<1$. If $\Omega_0=1$ then Ω is identically 1. Therefore, we can treat the universe as 'flat' (Einstein–de Sitter) during $\Omega_0^{-1}\ll 1+z\leqslant 1+z_{\mathrm{eq}}$. Here z_{eq} is the redshift at which the matter and radiation densities were equal. We also assume that the recombination occurred when the universe was matter dominated, that is $z_{\mathrm{eq}}>z_{\mathrm{rec}}$, where $z_{\mathrm{rec}}\sim 1500$ is the redshift at recombination.

Another assumption is that the significant perturbations were isothermal and were described by the spectrum

$$\delta\varrho/\varrho=[M/M_0(z)]^{-1/2-n/6} \tag{1}$$

where $M_0(z)$ defines a mass-scale at redshift z. $M_0(z=0)$ is estimated to be $10^{15}h^{-1}\,\Omega_0\,M_\odot$ (a mass of a supercluster). The power n is estimated empirically from the correlation function and from N-body simulations. $n=0$ corresponds to 'white noise' and $n=+1$ to the 'Zeldovich spectrum'. We have adopted the value $n=-1$ (Gott & Rees 1975). The main reason to ignore the adiabatic component is the constraint on the adiabatic matter perturbations to be of order 10^{-5} according to the background radiation (Uson & Wilkinson 1984). On the other hand, the matter perturbations at recombination should be of order 10^{-3} to form a bound system at present* (Weinberg 1972). By assuming isothermal perturbations we exclude a cut-off of the spectrum below the 'Silk mass' (Silk 1968). Hence, we can search for objects immediately after recombination. Until recombination the perturbations were damped by 'Thomson drag' and at $z<\Omega_0^{-1}$ (the inertial epoch) they froze. Hence, our discussion is about the period $\Omega_0^{-1}\ll 1+z<1+z_{\mathrm{rec}}$. The growth of linear perturbations in this era is proportional to $t^{2/3}$ (Weinberg 1972) and hence to the scale factor. Therefore, (1) can be written as

$$\delta\varrho/\varrho=[M/M_0(0)]^{-1/2-n/6}(1+z)^{-1}. \tag{2}$$

Each perturbed region with density larger than the critical density at recombination could form a bound system. We assume that the perturbed region was a spherical homogeneous non-rotating cloud. Opacity and magnetic field are ignored, but pressure is allowed for. The behaviour of the cloud's radius is analogous to that of the scale factor in Friedmann's solution for $\Omega>1$. Hence, the radius reaches a maximum value ('turn-around') and then recollapses, in isolation from the cosmological background. We discuss the collapse of each cloud (parametrized by its mass) from its turn-around radius.

We shall treat the universe as baryon-dominated with $\Omega_0=\Omega_{0,\,\mathrm{br}}=0.1$, where $\Omega_{0,\,\mathrm{br}}$ is the baryonic density parameter. We also take $h=1$. The mass fractions of primordial hydrogen and helium are taken to be $X=0.75$, $Y=0.25$, respectively, so that the mean molecular weight is $\mu=1.23$.

*The problems with the anisotropies in the adiabatic scenario go away if one has a neutrino-dominated universe.

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2 Cloud's interior

2.1 THE CHEMICAL REACTIONS

The post-recombination clouds were metal-poor, had low temperatures (10^2–10^3 K) and low densities (10^3–10^4 cm^{-3}). It was suggested that the main cooling mechanism is due to the H_2 molecule that is formed during the collapse (e.g. Takayanagy & Nishimura 1960; Hutchins 1976). This molecule's electronic ground state is $^1\Sigma_g^+$; it may be excited to rotational and vibrational levels as a result of collisions with hydrogen atoms. It will emit the photons in the infrared region of the spectrum and hence cool the cloud. Since H_2 is a homonuclear molecule, it has no dipole moment and that makes the cooling less efficient. It was suggested (Lepp & Shull 1984) that HD and LiH also contribute to the cooling because of their dipole moment. However, their abundances are so low compared to that of H_2 ($n_{LiH}/n_{H_2}=10^{-6.5}$; $n_{HD}/n_{H_2}=10^{-4.5}$ for $\Omega_0=1$, $\Omega_{0,br}=0.1$, $h=1$ at $z<60$) that the H_2 molecule remains the most important cooling agent.

Among the possible reactions of H_2 formation and destruction we have chosen those listed in Table 1. The reaction rates k_1, k_2, k_3, k_5 have units of cm^3 s^{-1} and k_4 is in s^{-1}. T_m and T_r indicate the matter and radiation temperatures, respectively. Our choice of reactions is based on arguments similar to those of Hutchins (1976) and on time-scales considerations. We have included reaction (4) which is important for $z>150$ and omitted the photo-disassociation of H_2 and photo-ionization of H since they are relevant for $z>1000$ only. We have neglected the reactions $H+H+H\rightarrow H_2+H$ and $H+H+H_2\rightarrow 2H_2$ (Palla *et al.* 1983) since they are relevant only for very high densities ($n>10^8$ cm^{-3}). We have adopted the reaction rate k_5 according to a new calculation of Lepp & Shull (1983). It has a lower value than k_5 of Hirasawa (1969) because it

Table 1. Formation and destruction reactions of H_2 molecule.

Reaction	*Rate*	References
(1) $H + e \rightarrow H^- + h\nu$	$k_1 = 1.83\times10^{-18}\, T_m^{0.8799}$	H
(2) $H + H^- \rightarrow H_2 + e$	$k_2 = 1.35\times10^{-9}$	H
(3) $H^+ + e \rightarrow H + h\nu$	$k_3 = 1.88\ 10^{-10}\, T_m^{-0.644}$	H
(4) $H^- + h\nu \rightarrow H + e$	$k_4 = 0.278\, T_r^2 \exp(-8{,}750/T_r)$	Hi
(5) $H + H_2 \rightarrow 3H$	$\log k_5 = \log k_H - \log(k_H/k_L)/(1+n_H/n_{cr})$	LS
	where:	
	$k_H = 3.52\times10^{-9} \exp(-4.39\times10^4/T_m)$	
	$k_L = \begin{cases} 6.11\times10^{-14} \exp(-2.93\times10^4/T_m) & T_m > 7390\ ^\circ K \\ 2.67\times10^{-15} \exp[-(6750/T_m)^2] & T_m \lesssim 7390\ ^\circ K \end{cases}$	
	$\log n_{cr} = 4.00 - 0.416\, y - 0.327\, y^2$	
	$y = \log T_m - 4.00$	

Notes: H=Hutchins (1976), Hi=Hirasawa (1969), LS=Lepp & Shull (1983). k_1, k_2, k_3, k_5 have units of cm^3 s^{-1} and k_4 is in s^{-1}.

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takes account of radiative decay of the upper vibrational levels. Lepp & Shull took into account collisional excitation and destruction from 15 vibrational levels.

The hydrogen number density n_H is much greater than the other densities and is therefore not altered by the chemical reactions. Its change is due to the contraction of the cloud

$$n_H(t)=n_{H,ta}[r_{ta}/r(t)]^3 \tag{3}$$

where $n_{H,ta}$ and r_{ta} are the density and radius at turn-around. Since $k_2 \gg k_1$ (and for $z>100$ $k_4 \gg k_1$) n_{H^-} is in equilibrium, so from the reactions of Table 1

$$n_{H^-}=k_1 n_H n_e/(k_2 n_H+k_4). \tag{4}$$

If we define $x_e=n_e/n_H$ we get by using (3) and (4)

$$\dot{x}_e=-k_3 n_H x_e^2. \tag{5}$$

The equation for $h_2=n_{H_2}/n_H$ is

$$\dot{h}_2=k_1 k_2 n_H^2 x_e/(k_2 n_H+k_4)-k_5 n_H h_2. \tag{6}$$

2.2 THE TEMPERATURE OF THE CLOUD

The law of temperature evolution as the cloud collapses is derived by using the first law of thermodynamics

$$dU=-P\,dV+dQ \tag{7}$$

where $U=3/2\,NkT$; $P=NkT/V$ are the internal energy and pressure of the monoatomic ideal gas. Q is heat input to the system. N is the total number of particles in the cloud, V is its volume and k is Boltzmann's constant. By using $V=NX\mu(r/r_{ta})^3/n_{H,ta}$ we get

$$\dot{T}=-2\,T\dot{r}/r-(2/3)X\mu\Lambda(r/r_{ta})^3/(kn_{H,ta}) \tag{8}$$

where Λ is the cooling function, in units of erg cm^{-3} s^{-1}. In adiabatic collapse ($\Lambda=0$) the solution is $T_{ad}=T_{ta}(r/r_{ta})^{-2}$. By using the new variables $x=r/r_{ta}$ and $f=T/T_{ad}$ we get the temperature equation

$$\dot{f}=-(2/3)X\mu\Lambda x^5/(kn_{H,ta}T_{ta}). \tag{9}$$

The cooling function Λ is usually written in the form

$$\Lambda=CT^{\alpha}n_H^{\beta}n_{H_2}. \tag{10}$$

For example, $C=2.37\times10^{-33}$; $\alpha=3.045$; $\beta=0.578$ in the range $100<T<3000$ K and $10^{-1}<n_H<10^4$ cm^{-3} (Hutchins 1976). We have adopted a more detailed form by Lepp & Shull (1983). They computed dissociation and radiation of H_2 as a result of H_2–H and H_2–H_2 collisions, by taking into account four vibrational levels, each with 21 rotational levels. Their expression, neglecting H_2–H_2 collisions, is

$$\Lambda=n_{H_2}[L_{RH}/(1+L_{RH}/L_{RL})+L_{VH}/(1+L_{VH}/L_{VL})] \tag{11}$$

where L_{RH}, L_{RL}, L_{VH}, L_{VL} are the cooling coefficients at high and low densities, in units of erg s^{-1}

$$L_{VH}=1.10\times10^{-13}\exp(-6744/T)$$

$$L_{VL}=8.18\times10^{-13}\,n_H\times\begin{cases}1.0\times10^{-12}\,T^{1/2}\exp(-1000/T) & T>1635\text{ K}\\ 1.4\times10^{-13}\exp[(T/125)-(T/577)^2] & T\leqslant1635\text{ K}\end{cases}$$

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$$L_{RH}=\begin{cases}3.90\times10^{-9}\exp(-6118/T) & T>1087\,\mathrm{K}\\ \mathrm{dex}[-19.24+0.474\,y-1.247\,y^2] & T\leqslant 1087\,\mathrm{K}\end{cases}$$

$$L_{RL}=(n_{H_2}^{0.77}+1.2\,n_H^{0.77})\times\begin{cases}1.38\times10^{-22}\exp(-9234/T) & T>4031\,\mathrm{K}\\ \mathrm{dex}[-22.90-0.553\,y-1.148\,y^2] & T\leqslant 4031\,\mathrm{K}\end{cases}$$

where $y=\log(T)-4$.

2.3 THE CLOUD'S RADIUS

By contrast to other authors we do not assume *a priori* that the cloud's collapse is a free-fall. We build a simple equation that takes into account the pressure. The energy equation of the homogeneous cloud of mass M is

$$\dot{E}=-\Lambda V \tag{12}$$

where $E=-3/5\,GM^2/r+3/10\,M(\dot{r})^2+3/2\,NkT$ is the sum of the potential, kinetic and internal energies. By using (8) and $N=M/(\mu m_H)$, where m_H is the mass of hydrogen atom, we get

$$\ddot{r}=5k/(\mu m_H)T/r-GM/r^2. \tag{13}$$

The first term on the right-hand side describes the influence of the pressure.

In adiabatic collapse $T\propto r^{-2}$ and we observe that in this case (13) is analogous to the equation of motion of the Kepler's problem (the pressure term acts like the centrifugal acceleration). For initial conditions $r(0)=r_{ta}$; $\dot{r}(0)=0$ we get the parametric solution (*cf.* Landau & Lifshitz 1976)

$$r=a[1+e\cos(\zeta)]$$

$$t=(GM)^{-1/2}a^{3/2}[\zeta+e\sin(\zeta)] \tag{14}$$

where $0\leqslant\zeta\leqslant2\pi$; $a=(r_{ta}/2)(1+U_{ta}/W_{ta})^{-1}$; the 'eccentricity' $e=1+2\,U_{ta}/W_{ta}$, where U_{ta}, W_{ta} are the internal and potential energies at the turn-around. The case $e=1$ ('parabolic Keplerian orbit') corresponds to a free-fall collapse, while $e=0$ ('circular orbit') simply describes a virialized system. The ratio of the time it takes the cloud to fall from the 'aphelion' to the 'perihelion' to the free-fall time of the same cloud is $(1+U_{ta}/W_{ta})^{-3/2}$.

From (13) we can also define the Jeans mass by the requirement $\ddot{r}=0$

$$M_J=2.38\times10^{23}(T/\mu)^{3/2}\varrho^{-1/2}\,\mathrm{g}. \tag{15}$$

The condition for a collapse to begin is that the cloud's mass M be greater than M_J.

We can rewrite (13) with the help of the definitions $x=r/r_{ta}$; $f=T/T_{ad}=(T/T_{ta})x^2$

$$\ddot{x}=5kT_{ta}(\mu m_H r_{ta}^2)^{-1}fx^{-3}-GM/r_{ta}^{-3}x^{-2}. \tag{16}$$

Probably, in a realistic pressure supported collapse there would be a collision of shells (unlike in a free-fall collapse). A density contrast may build up in the core and shock-waves could develop. If this happened H_2 might cool down the shock-waves (MacLow & Shull 1985). We can estimate the post-shock temperature T_s by comparing the pre-shock kinetic energy of an atom in free-fall [$m_H\dot{r}^2/2\approx GMm_H/r_{ta}$, where r_{ta} is given by (19)] to the post-shock thermal energy $(3/2)\,kT_s$. This simple estimate is correct both for a free-fall and pressure supported collapse. In a free-fall collapse the pre-shock gas is cold but has a high velocity, while in a pressure supported collapse the pre-shock velocity is low, but the gas is hot. For a cloud of $M=10^6\,M_\odot$ one gets $T_s\approx18\,000$ K, which is an intermediate value between the temperatures at which H_2 cooling and L_α cooling are important. Hence, a cooling of the shocks is possible.

Other factors which we have neglected could act against the development of inhomogeneities; e.g. rotation and magnetic field (the latter would increase the Jeans mass).

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The above qualitative arguments show that although the physics of the collapsing cloud is much more complicated than our simple model, equation (13) might describe the cloud's behaviour in a more realistic way than a free-fall model. In particular, it may describe the motion of the outer shells of the cloud.

The equations (5), (6), (9), (16) form a complete set of coupled differential equations that describes the cloud's interior. In the next section we shall define the initial conditions for the above equations, according to the cosmological background.

3 The cosmological initial conditions

3.1 THE MATTER DENSITY

The matter density in a matter dominated universe is given by

$$\varrho_m=\Omega_0\varrho_{cr,0}(1+z)^3=1.88\times10^{-29}\,\Omega_0h^2(1+z)^3\,\mathrm{g\,cm^{-3}} \tag{17}$$

where $\varrho_{cr,0}$ is the critical density at present.

The density of a mass at 'turn-around' time is

$$\varrho_{ta}=(3\pi/4)^2 1.88\times10^{-29}\,\Omega_{0,br}h^2(1+z)^3\,\mathrm{g\,cm^{-3}}. \tag{18}$$

The factor $(3\pi/4)^2$ is the over-density of the cloud at turn-around compared to the background (Peebles 1980). It can also be obtained by the requirement that the free-fall time-scale is less than or equal to the expansion time-scale. We have replaced Ω_0 by $\Omega_{0,br}$ according to our assumption that all the matter is baryonic. The number density of hydrogen atoms at turn-around is given by $n_{H,ta}=X\varrho_{ta}/m_H$. The radius at turn-around is (by using 18)

$$r_{ta}=[M/(4\pi\varrho_{ta}/3)]^{1/3}=1.66\times10^{20}(1+z_{ta})^{-1}[(M/M_\odot)/(\Omega_{0,br}h^2)]^{1/3}\,\mathrm{cm}. \tag{19}$$

3.2 THE ELECTRON DENSITY

The residual electrons after recombination play an important role in the H_2 formation [via reaction (1)] and in coupling the matter and radiation. We adopt the expression of Sunyaev & Zeldovich (1970, 1980) that takes into account the two-photon decay from the 2S to 1S levels of the hydrogen atom.

$$x_e(z)\equiv n_e/(n_e+n_H)=\begin{cases}A/(zh\Omega_0^{1/2})\exp(-B/z) & 800<z<1500\\ C/(h\Omega_0^{1/2}) & z\leqslant800.\end{cases} \tag{20}$$

We take the constants as: $A=1.8\times10^6$; $B=1.46\times10^4$; $C=2.7\times10^{-5}$.

3.3 THE RADIATION AND MATTER TEMPERATURES

The radiation temperature is given by

$$T_r=2.7(1+z). \tag{21}$$

If the decoupling of matter and radiation were complete the matter temperature for $z\leqslant z_{eq}$ would be given by

$$T_m=T_{rec}[(1+z)/(1+z_{rec})]^2=1.8\times10^{-3}(1+z)^2 \tag{22}$$

where we have used $T_{rec}=2.7(1+z_{rec})=4050\,\mathrm{K}$.

However, there is a coupling of matter and radiation due to the Thomson scattering of the

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photons by the free electrons surviving after recombination, so T_m will be different from (22). This temperature is important; it plays the role of the initial cloud's temperature which determines the Jeans mass, the reaction rates and the cooling function. Following Peebles (1968) the equation of the matter temperature in an expanding universe is

$$\dot{T}_m=(8/3)\sigma_T a T_r^4 x_e(T_r-T_m)/(m_e c)-2T_m\dot{R}/R \tag{23}$$

where $\sigma_T=6.652\times10^{-25}$ cm^2 is the Thomson cross-section, $a=7.564\times10^{-15}$ erg cm^{-3} K^{-4} and R is the scale factor. The first term on the right-hand side describes the heat transfer from the photons to the electrons (the atoms, photons and electrons are in thermodynamic equilibrium). The second term shows the cooling due to expansion [solution (22)]. We shall change the independent variable (23) from time to redshift. In a 'flat' universe for redshift $1+z\gg\Omega_0^{-1}$ the time is expressed as

$$t=2/(3\mathrm{H})=2.06\times10^{17}h^{-1}\Omega_0^{-1/2}(1+z)^{-3/2}\ \mathrm{s}. \tag{24}$$

By introducing (24) into (23) we get

$$dT_m/d(1+z)=-8.06\times10^{-3}h^{-2}\Omega_0^{-1}c_e(1+z)(1+z)^{3/2}[2.7(1+z)-T_m]+2T_m/(1+z) \tag{25}$$

where $c_e(z+1)=h\Omega_0^{1/2}x_e(z+1)$ and $x_e(z)$ is given by (20).

The initial condition is $T_m(1+z_{rec}=1500)=2.7\times1500=4050$ K. We shall adopt an approximate solution of the form

$$T_m\cong0.1(1+z)^{1.5} \tag{26}$$

for the range $100\leqslant1+z\leqslant1000$ and the value of $h^2\Omega_0=0.1$. The adiabatic temperature of monoatomic gas is proportional to $\varrho^{2/3}$ and hence (26) should be multiplied by the factor $(3\pi/4)^{4/3}\sim\pi$ at turn-around.

3.4 THE REDSHIFT AT TURN-AROUND

We shall relate the mass of each cloud to the redshift at its turn-around. Gunn & Gott (1972) calculated the behaviour of a spherical shell with average density $\bar{\varrho}_i$ greater than the critical density ϱ_{ci} at time t_i. The total time from t_i to recollapse (assuming $t_i\ll t_c$) is

$$t_c=2t_{ta}\approx(\pi/\mathrm{H}_i)[\varrho_{ci}/(\bar{\varrho}_i-\varrho_{ci})]^{3/2}. \tag{27}$$

If we identify $[(\bar{\varrho}_i-\varrho_{ci})/\varrho_{ci}]$ with $(\delta\varrho/\varrho)_i$ of (2) and use (24) we get

$$1+z_{ta}=(3\pi/4)^{-2/3}[M/M_0(0)]^{-1/2-n/6}. \tag{28}$$

We see that the only required initial parameter of the cloud is its mass. From that we can calculate the redshift at turn-around and according to that all other initial values.

4 The numerical solutions

Before solving the full set of equations we shall write an approximate equation for the temperature as a function of the cloud's radius. We neglect k_3 (recombination), k_5 (destruction of H_2, which is important in the last stages of collapse) and k_4 (photo-disassociation of H^-, which is important at high redshift). We also neglect the pressure and assume a free-fall collapse (a relevant assumption for the first stages of collapse). The free-fall velocity of the cloud is therefore $\dot{x}=-\pi/(2t_{ff})\sqrt{(1-x)/x}$ where $x=r/r_{ta}$ and t_{ff} is the free-fall time. For simplicity we write k_1 as $k_1=k_0T$ cm^3 s^{-1} where $k_0=1.83\times10^{-18}$ and $\Lambda=CT^\alpha n_H^\beta n_{H_2}$ with $C=2.37\times10^{-33}$, $\beta=0.578$,

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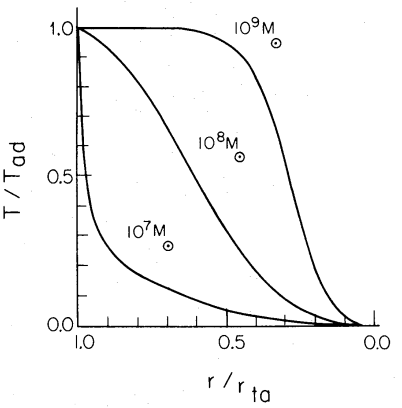


Figure 1. The numerical solution of equation (29), for three values of cloud's mass. The temperature (relative to the temperature in adiabatic collapse) is shown versus the radius (relative to the initial radius). The cosmological parameters are: $h=1$, $\Omega_0=\Omega_{0,\mathrm{br}}=0.1$.

$\alpha=3.045$ (Hutchins 1976). We are left with equations (6) and (9) and their combination gives

$$f''=\alpha(f')^2/f+\{1/[2(1-x)]-p/x\}f'-Kf^{\alpha+1}/[x^{p+4.5}(1-x)] \tag{29}$$

where $p=2\alpha+3\beta-2.5$ and K depends on the initial conditions

$$K=k_0CX^2\mu/(4\pi Gm_{\mathrm{H}}k)\,T_{\mathrm{ta}}^{\alpha}n_{\mathrm{H,ta}}^{\beta}x_{\mathrm{e,ta}}.$$

The initial conditions at $x=1$ are $f=1$, $f'=0$. The derivatives are with respect to x.

We did not find an analytic solution for this second order non-linear equation. We solved it numerically (by writing it as a set of two first order differential equations) with the subroutine DVERK of IMSL. The solution of (29) is shown in Fig. 1 for three masses. The main feature is that there is a strong decline in temperature from a certain radius. The solution resembles the Fermi function

$$f(x)=\{1+\exp[(x_0-x)/\Delta]\}^{-1}. \tag{30}$$

Here x_0 is the value of x in which $f=1/2$. x_0 and Δ depend on the redshift z_{ta} e.g., for $M=10^9\,M_\odot$, $x_0=0.3$ and $\Delta=0.06$.

Next, we have solved the full set of equations [(5), (6), (9), (16)] by using subroutine DGEAR of IMSL (which is useful for stiff equations). We have used the reaction rates of Table 1 and the cooling function (11). For the initial conditions we have used the relations of Section 4: $x_{\mathrm{e}}(0)$ according to (20), $T(0)=T_{\mathrm{ad}}$ according to (26), $r(0)=r_{\mathrm{ta}}$ according to (19), $\dot{r}(0)=0$ and $h_2(0)=0$. All the initial conditions are determined from the cloud's mass via (28). The validity of the numerical calculation was checked by comparing some special cases (free-fall, adiabatic collapse) to analytic solutions. The agreement is to about 2 per cent.

Fig. 2 depicts x_{e}, h_2, T/T_{ad}, r/r_{ta} versus the time (in units of free-fall time) for a cloud of mass $10^6\,M_\odot$ (for the cosmological parameters $h=1$, $\Omega_0=\Omega_{0,\mathrm{br}}=0.1$). The Jeans mass is also shown. The initial values in such a cloud are, $z_{\mathrm{ta}}=570$, $n_{\mathrm{H,ta}}=870\,\mathrm{cm}^{-3}$, $r_{\mathrm{ta}}=6\times10^{19}\,\mathrm{cm}$, $x_{\mathrm{e}}=8\times10^{-5}$, $T_{\mathrm{ta}}=3700\,\mathrm{K}$. The free-fall time is $t_{\mathrm{ff}}=5\times10^{13}\,\mathrm{s}$. At the final calculated radius the values are: $\varrho_{\mathrm{f}}=7\times10^{-18}\,\mathrm{g\,cm}^{-3}$, $x_{\mathrm{e,f}}=2\times10^{-7}$, $h_{2,\mathrm{f}}=2\times10^{-4}$, $T_{\mathrm{f}}=370\,\mathrm{K}$, $M_J=230\,M_\odot$. The maximum value reached by the temperature during the collapse is 21 600 K. This indicates the importance of cooling by vibrational transitions (in such high temperature L_α cooling should also be taken into account). An important feature is the oscillations of the cloud's radius due to the pressure. As a result, the cloud collapsed after $3\,t_{\mathrm{ff}}$. It collapsed at $1+z=120$ instead of the free-fall value $1+z=220$.

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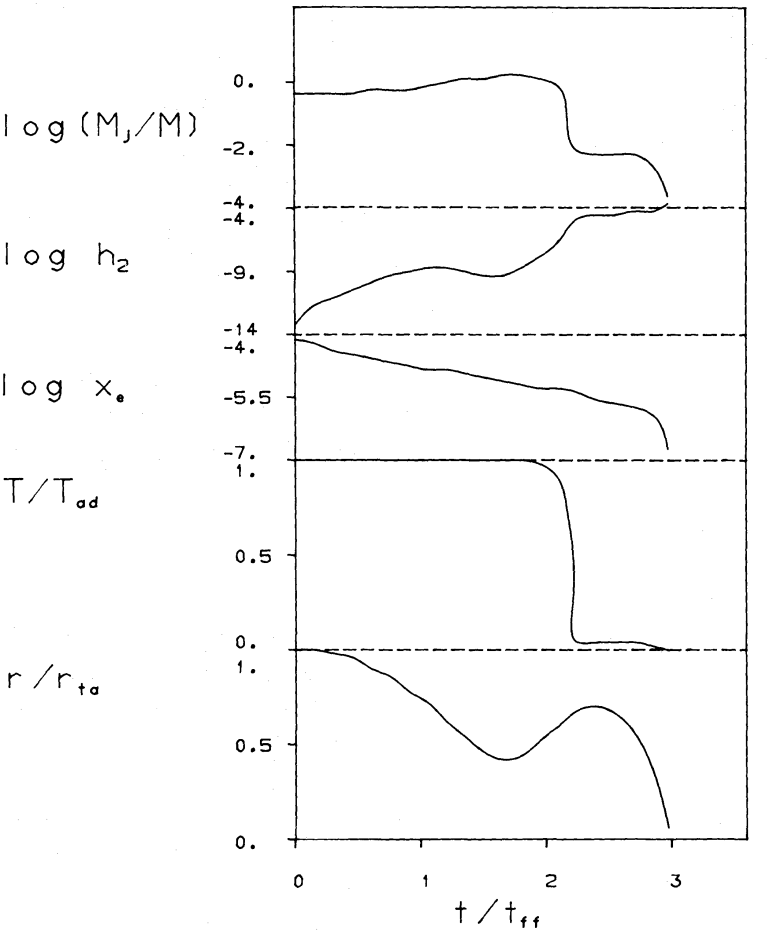


Figure 2. The physical quantities in a collapsing cloud of $10^6 M_\odot$ as a function of time (in units of free-fall time). The shown quantities are: the radius (relative to the initial radius), the temperature (relative to the adiabatic temperature), electron and H_2 abundances and the Jeans mass (relative to the cloud's mass). The cosmological parameters are: $h=1$, $\Omega_0=\Omega_{0,\mathrm{br}}=0.1$.

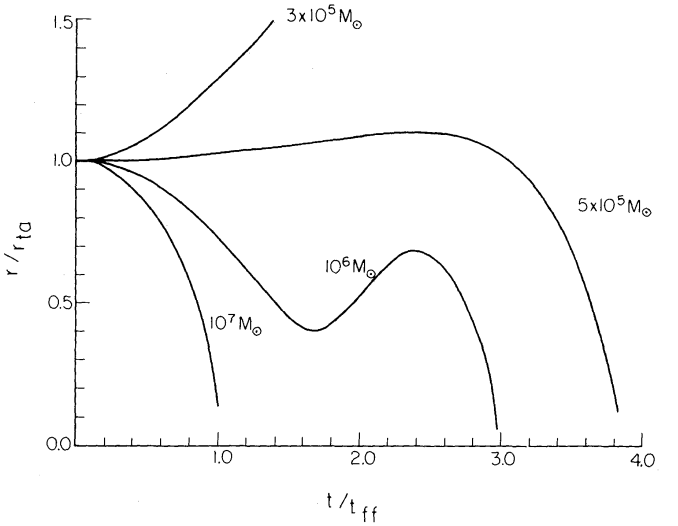


Figure 3. The cloud's radius (relative to the initial radius) for various masses as a function of time (in units of free-fall time of each cloud). The cosmological parameters are: $h=1$, $\Omega_0=\Omega_{0,\mathrm{br}}=0.1$.

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Fig. 3 shows the behaviour of the cloud's radius for various masses. It seems that there are four modes of behaviour

(i) $M \leqslant 3\times10^5\,M_\odot$ an expansion of cloud.

(ii) $5\times10^5\,M_\odot \geqslant M > 3\times10^5\,M_\odot$: an expansion of the cloud (according to Jeans criterion) and then a collapse, due to H_2 formation during the expansion.

(iii) $5\times10^6\,M_\odot > M > 5\times10^5\,M_\odot$: oscillations, under the influence of pressure.

(iv) $M \geqslant 5\times10^6\,M_\odot$: an immediate collapse (pressure is unimportant).

We also see that masses of $5\times10^5-10^7 M_\odot$ completed their collapse at $z=100-140$ (instead of the free-fall value $z=100-270$).

We also considered the case $h_2(0)=10^{-6}$, due to H_2 formation during expansion of the universe. We got similar results, but the four modes of behaviour occurred at smaller mass, e.g., $10^6\,M_\odot$ collapses within $1.1\,t_{\rm ff}$. By introducing k_5 of Hutchins (1976) we got a collapse of $10^6\,M_\odot$ only after $4.5\,t_{\rm ff}$.

5 Discussion

We have explored the influence of the pressure on the behaviour of the cloud's radius and on the epoch of collapse. The collapse stops when opacity is becoming important or when the hydrogen molecules are destructed. The cloud may fragment into smaller masses of order of the minimal Jeans mass during collapse [the calculated masses of fragments are in the wide range $0.1-200\,M_\odot$ (Hutchins 1976; Silk 1977; Carlberg 1981; Palla *et al.* 1983)].

We suggest that the different modes of radius behaviour could give rise to different ways of fragmentation and therefore to different families of objects with a similar mass. By analogy, in 'young' populations there are star systems with similar masses which are different morphologically; i.e. the globular clusters and the dwarf elliptical galaxies (Kormendy 1984).

According to our calculations the mass of the first bound systems is about $10^6\,M_\odot$, in agreement to the results of Peebles & Dicke (1968) and Carr & Rees (1984).

A more rigorous treatment of the problem should describe the cloud in its phase of expansion, before the turn-around. Other important factors are non-spherical collapse, non-homogeneous density distribution, rotation, magnetic field and opacity.

Acknowledgments

I thank Professor Jacob Bekenstein for his support and advice during the development of this work. I also acknowledge helpful discussions with Professor Martin Rees and Dr Sasha Kashlinsky.

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